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# Predictive Analytics for Customer Behavior
# Nyeras Data Science Project
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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
import random

names = ["Aarav", "Ananya", "Rahul", "Priya", "Kiran", "Sneha", "Arjun", "Divya", "Rohi"]
cities = ["Hyderabad", "Bangalore", "Chennai", "Mumbai", "Delhi", "Pune", "Kolkata", ""]
genders = ["Male", "Female"]

data = []

for i in range(1, 501):
    age = random.randint(18, 60)
    income = random.randint(20000, 150000)
    purchase_count = random.randint(1, 50)
    total_spend = purchase_count * random.randint(500, 5000)
    last_purchase_days = random.randint(1, 365)

    churn = 1 if (last_purchase_days > 180 and purchase_count < 10) else 0

    data.append([
        i,
        random.choice(names),
        age,
        random.choice(genders),
        random.choice(cities),
        income,
        purchase_count,
        total_spend,
        last_purchase_days,
        churn
    ])

df = pd.DataFrame(data, columns=[
    "Customer_ID",
    "Name",
    "Age",
    "Gender",
    "City",
    "Income",
    "Purchase_Count",
    "Total_Spend",
```

```

    "Last_Purchase_Days",
    "Churn"
])

df.head()

```

	Customer_ID	Name	Age	Gender	City	Income	Purchase_Count	Total_Spend
0	1	Kiran	37	Female	Chennai	40389	17	2866
1	2	Divya	38	Male	Pune	136441	38	7714
2	3	Priya	50	Female	Mumbai	100223	38	9066
3	4	Rohit	56	Male	Delhi	122157	2	451
4	5	Priya	43	Male	Hyderabad	106974	17	8095

```

df.to_csv("customer_behavior.csv", index=False)
print("Dataset saved successfully!")

```

Dataset saved successfully!

```
df.shape
```

(500, 10)

```
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Customer_ID           500 non-null   int64
1   Name                  500 non-null   object
2   Age                   500 non-null   int64
3   Gender                500 non-null   object
4   City                  500 non-null   object
5   Income                500 non-null   int64
6   Purchase_Count        500 non-null   int64
7   Total_Spend           500 non-null   int64
8   Last_Purchase_Days    500 non-null   int64
9   Churn                 500 non-null   int64
dtypes: int64(7), object(3)
memory usage: 39.2+ KB

```

```
df.head(10)
```

	Customer_ID	Name	Age	Gender	City	Income	Purchase_Count	Total_Sp
0	1	Kiran	37	Female	Chennai	40389	17	28
1	2	Divya	38	Male	Pune	136441	38	77
2	3	Priya	50	Female	Mumbai	100223	38	90
3	4	Rohit	56	Male	Delhi	122157	2	4
4	5	Priya	43	Male	Hyderabad	106974	17	80
5	6	Kiran	33	Female	Mumbai	142720	22	32
6	7	Priya	37	Male	Delhi	87381	42	113
7	8	Ananya	24	Female	Mumbai	97277	11	30
8	9	Sneha	25	Female	Chennai	134333	23	84
9	10	Rohit	46	Male	Delhi	123599	12	58

```
df.describe()
```

	Customer_ID	Age	Income	Purchase_Count	Total_Spend	Last_
count	500.000000	500.000000	500.000000	500.000000	500.000000	
mean	250.500000	38.958000	86914.168000	24.584000	66894.532000	
std	144.481833	12.036860	37909.423850	13.949054	52032.667279	
min	1.000000	18.000000	20667.000000	1.000000	503.000000	
25%	125.750000	29.000000	55442.500000	13.750000	24093.500000	
50%	250.500000	39.000000	89268.000000	23.000000	56310.500000	
75%	375.250000	49.000000	119588.500000	36.000000	94806.000000	
max	500.000000	60.000000	149338.000000	50.000000	238368.000000	

```
df.isnull().sum()
```

	0
Customer_ID	0
Name	0
Age	0
Gender	0
City	0
Income	0
Purchase_Count	0
Total_Spend	0
Last_Purchase_Days	0
Churn	0

dtype: int64

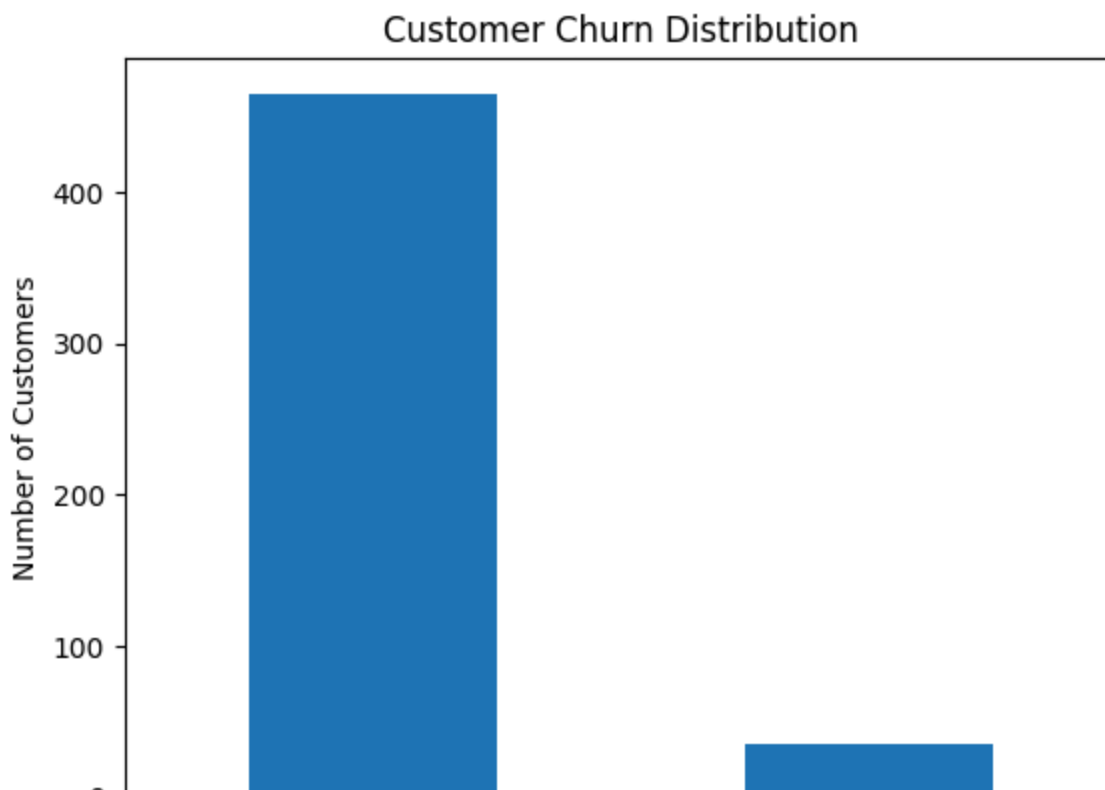
```
df["Churn"].value_counts()
```

	count
Churn	
0	465
1	35

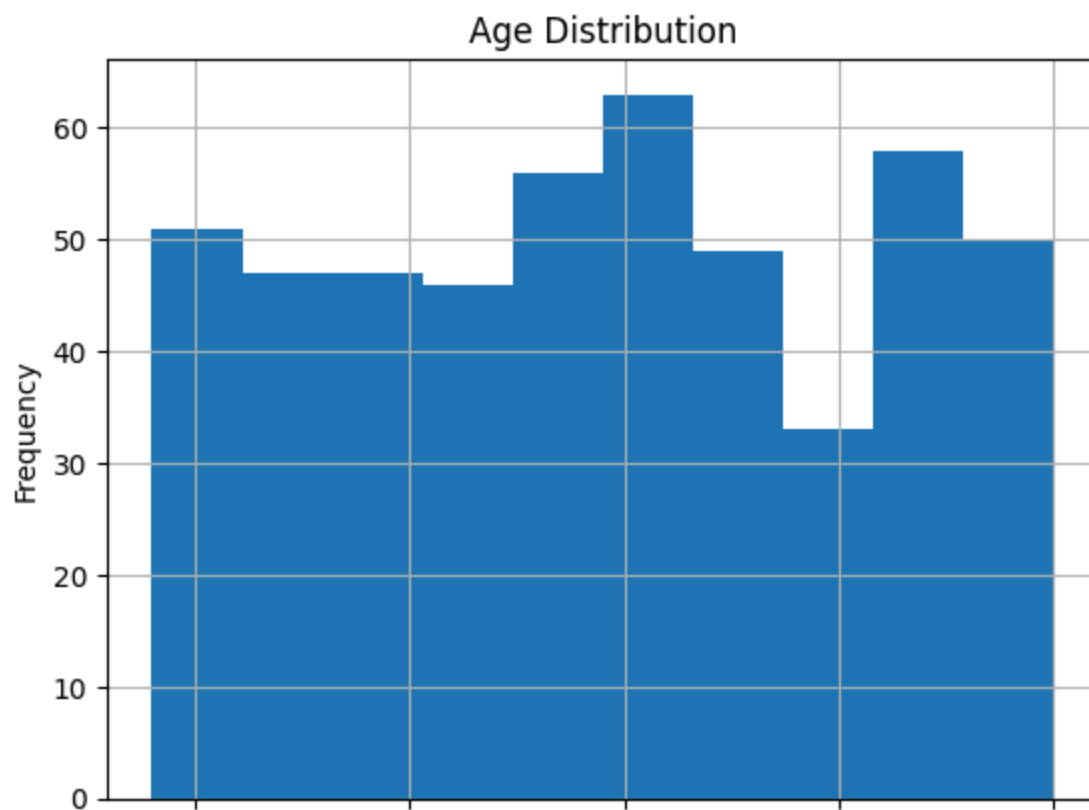
dtype: int64

```
import matplotlib.pyplot as plt

df["Churn"].value_counts().plot(kind="bar")
plt.title("Customer Churn Distribution")
plt.xlabel("Churn")
plt.ylabel("Number of Customers")
plt.show()
```



```
df["Age"].hist(bins=10)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```



```

from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()

df["Gender"] = le.fit_transform(df["Gender"])
df["City"] = le.fit_transform(df["City"])
df["Name"] = le.fit_transform(df["Name"])

df.head()

```

	Customer_ID	Name	Age	Gender	City	Income	Purchase_Count	Total_Spend	La:
0	1	4	37	0	1	40389	17	28662	
1	2	3	38	1	6	136441	38	77140	
2	3	6	50	0	5	100223	38	90668	
3	4	8	56	1	2	122157	2	4510	
4	5	6	43	1	3	106974	17	80954	

```

x = df.drop("Churn", axis=1)
y = df["Churn"]

print(x.shape)
print(y.shape)

```

```

(500, 9)
(500,)

```

```

from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(
    x,
    y,
    test_size=0.2,
    random_state=42
)
print(x_train.shape)
print(x_test.shape)

```

```

(400, 9)
(100, 9)

```

```
from sklearn.ensemble import RandomForestClassifier
```

```
model = RandomForestClassifier(random_state=42)
model.fit(x_train, y_train)
```

```
print("Model trained successfully!")
```

```
Model trained successfully!
```

```
y_pred = model.predict(x_test)
```

```
print(y_pred[:10])
```

```
[1 0 0 0 0 0 0 0 0 0]
```

```
from sklearn.metrics import accuracy_score
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Accuracy:", accuracy)
```

```
Accuracy: 1.0
```

```
from sklearn.metrics import confusion_matrix
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
print(cm)
```

```
[[95  0]
 [ 0  5]]
```

```
import pandas as pd
```

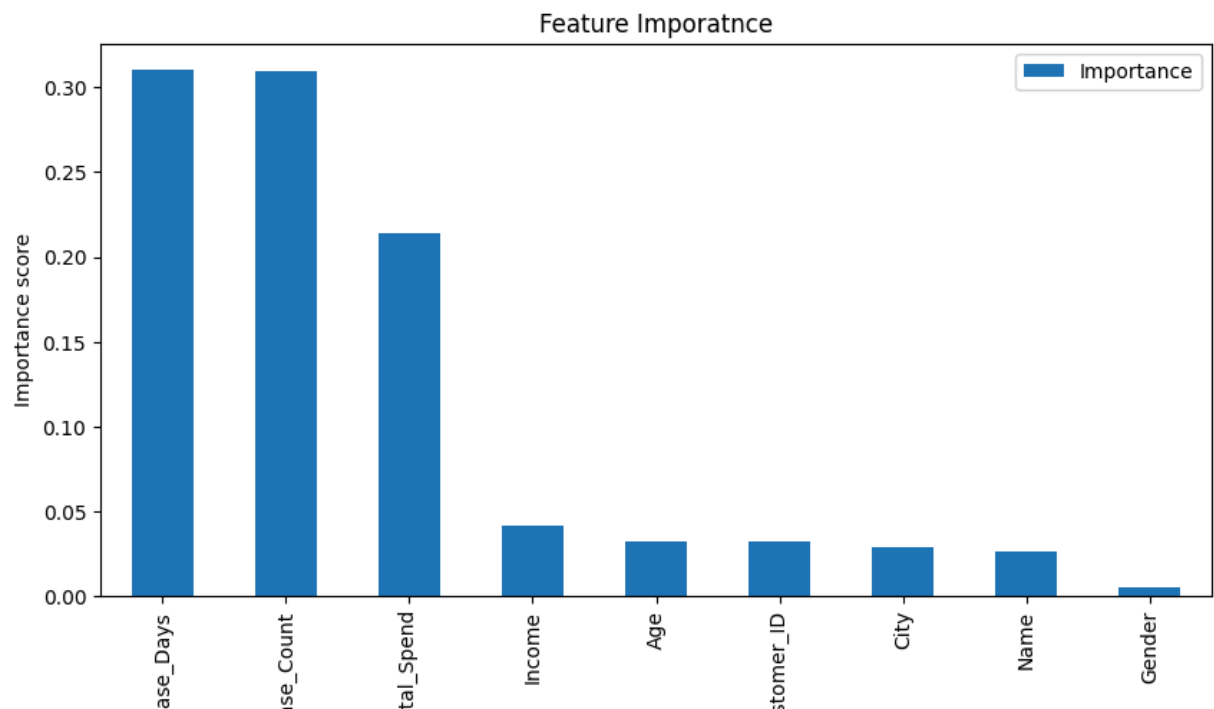
```
importance = pd.DataFrame({
    "Feature": x.columns,
    "Importance": model.feature_importances_
})
```

```
importance = importance.sort_values(by="Importance", ascending=False)
```

```
print(importance)
```

	Feature	Importance
8	Last_Purchase_Days	0.310220
6	Purchase_Count	0.310033
7	Total_Spend	0.213810
5	Income	0.041700
2	Age	0.032341
0	Customer_ID	0.031989
4	City	0.028659
1	Name	0.026091
3	Gender	0.005157

```
importance.plot(  
    x="Feature",  
    y="Importance",  
    kind="bar",  
    figsize=(10,5)  
)  
plt.title("Feature Importatnce")  
plt.ylabel("Importance score")  
plt.show()
```



```
new_customer = pd.DataFrame([  
    {"Customer_ID": 501,  
     "Name": 1,  
     "Age": 30,  
     "Gender": 1,  
     "City": 2,  
     "Income": 60000,  
     "Purchase_Count": 8,  
     "Total_Spend": 25000,  
     "Last_Purchase_Days": 220  
}])  
  
prediction = model.predict(new_customer)
```